

Assessment of the influence of corrosion on the structural performance of a steel truss railway bridge

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Abstract. Steel railway infrastructures located in marine environments are particularly vulnerable to corrosion phenomena, which may progressively impair the efficiency of structural connections and boundary conditions, affecting both the static behaviour and the global dynamic response. In such contexts, the assessment of the effectiveness of extraordinary maintenance interventions requires investigation protocols capable of capturing structural changes that are not readily detectable through visual inspections alone.

This work presents an integrated experimental approach for the assessment of the structural performance of steel railway bridges, based on the comparison between testing campaigns carried out before and after maintenance interventions. The protocol combines static load tests, aimed at evaluating structural deformations and stress states, with dynamic characterization tests performed through Operational and Experimental Modal Analysis, allowing the identification of natural frequencies, mode shapes and damping ratios.

The application to the Forgia railway bridge, consisting of steel truss girders affected by an advanced state of corrosion, highlights how restoration interventions on bolted joints and steel surfaces produce measurable effects not only on the static response, but also on boundary conditions and energy dissipation capacity. The results show that dynamic analysis represents a particularly sensitive tool for interpreting the evolution of structural performance and for defining a reliable reference baseline for long-term monitoring.

Keywords: Corrosion, Steel truss railway bridges, Structural dynamics, Experimental assessment, Operational Modal Analysis.

1 Introduction

Steel railway bridges constitute a significant part of existing infrastructure assets. Many of these structures feature efficient, lightweight designs where overall behavior depends heavily on the efficiency of bolted connections, bracing systems, and boundary conditions. When located near the coast, the combined effects of humidity, salt spray,

and hygrothermal cycles accelerate corrosion, complicating the assessment of residual safety [1,2].

Corrosion causes more than just a localized reduction in cross-sectional area. In truss bridges, it can alter joint behavior, reduce or modify relative slip between connected components, and consequently influence both static and global dynamic responses. Therefore, while visual inspection is essential, it is not always sufficient to quantify the impact of deterioration or the effectiveness of major maintenance interventions.

In this context, a coordinated approach combining static and dynamic testing enables a more comprehensive assessment. Load tests provide direct data on deflections, deformations, and stresses induced by controlled loads. Conversely, dynamic tests—conducted using Operational Modal Analysis (OMA) and Experimental Modal Analysis (EMA)—allow for the identification of natural frequencies, mode shapes, and damping coefficients; these parameters are particularly sensitive to changes in stiffness, boundary conditions, and energy dissipation [3,4].

This paper describes the application of this approach to the Forgia railway bridge, located on the Trapani-Palermo line. The structure, which exhibited widespread corrosion at bolted joints, underwent an experimental testing campaign both before and after the intervention. Comparing the results makes it possible to interpret the effects of the restoration interventions and to establish an experimental baseline for potential future monitoring.

2 Case study and state of deterioration

2.1 Description of the work



Fig. 1. Side view of the Forgia Bridge.

The Forgia Bridge is a railway structure on the Trapani-Palermo line, located within the municipality of Balestrate, near the mouth of the stream of the same name and a short distance from the shoreline. Its location results in a harsh environmental exposure,

characterized by the presence of salt spray and conditions conducive to the initiation and propagation of corrosion on the metal components.

The structure consists of metal trusses featuring a Warren configuration with a bottom-chord track layout and upper bracing. The main elements are constructed from rolled sections connected by bolted joints—a common design solution for metal railway bridges of that era [1]. The span between the bearing centerlines is 39.93 m; the spacing between the two trusses is 5.20 m, while the truss height is 6.90 m.

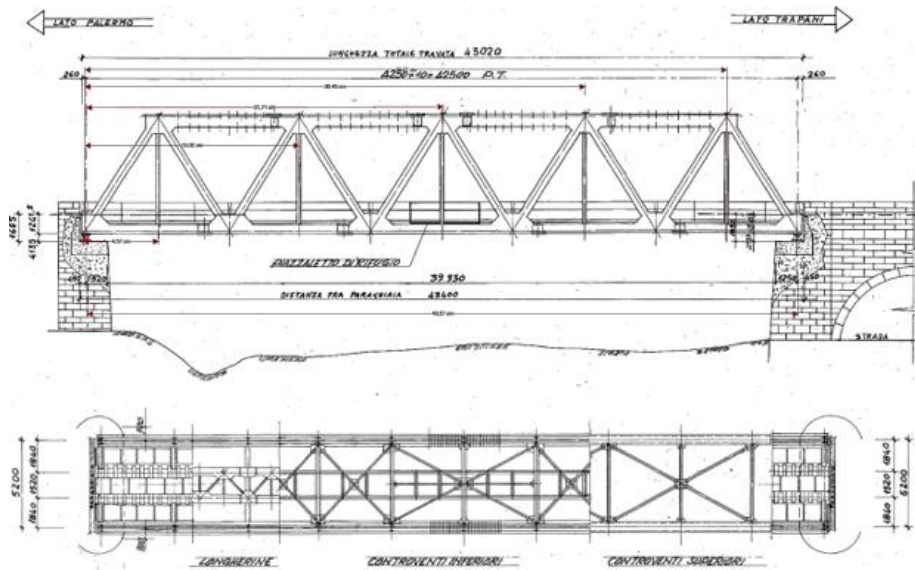


Fig. 2. General layout of the structure and main geometric dimensions.

Table 1. Main geometric and structural characteristics of the Forgia Bridge.

Parameter	Value
Structural scheme	Warren-type metal truss with a bottom chord deck
Clear span between bearing centerlines	39.93 m
Center-to-center distance between the two spans	5.20 m
Height of the truss girders	6.90 m
Connection type	Bolted connections on rolled sections

2.2 Corrosive degradation and maintenance intervention

The deterioration observed prior to the work was primarily attributable to the corrosion of metal surfaces and, more notably, of the bolted connections. In some instances, oxidation had significantly affected the bolts, potentially reducing their load-bearing cross-section and consequently compromising the local efficiency of the connection. This phenomenon is particularly critical in a truss bridge, as the connections facilitate the transfer of axial forces between members and maintain the system's overall structural stiffness.

The major maintenance project involved replacing the bolts, abrasive blasting the metal components, and applying a protective coating. The objective was not only to halt the surface deterioration process but also to restore connection conditions that aligned with the structure's original configuration, thereby enhancing its durability and operational performance.



Fig. 3. State of corrosive degradation observed on bolted joints and metal surfaces.

3 Influence of corrosion on structural behaviour

In metal elements exposed to marine environments, corrosion can produce diverse and sometimes competing effects. On one hand, it causes material loss, thereby reducing local load-bearing capacity; on the other, it can alter connection behaviour through localized oxidation between plates, splice plates, and bolts.

In bolted joints, oxide formation can increase local friction, partially restrict relative slip, and create boundary conditions that differ from those envisaged in the original design.

These aspects are particularly significant for truss bridges, where force transfer relies on a system of members and connections. Even a localized change in joint efficiency can affect the apparent stiffness of the entire truss, its deformation under load, and experimentally identified dynamic parameters. Consequently, any change in natural frequencies following maintenance cannot be automatically interpreted as a simple increase or decrease in structural safety.

Dynamic analysis is particularly useful in this context. Frequencies, mode shapes, and damping characteristics not only describe global stiffness but also reflect the system's boundary conditions and energy dissipation capabilities. Comparing pre- and post-intervention data thus makes it possible to assess whether restoring connections

yields a more consistent response, even when frequency changes are minor or appear counterintuitive.

4 Pre- and post-intervention experimental protocol

4.1 Investigation programme

The experimental protocol was organized into two distinct campaigns. The first, conducted in July 2021, documented the condition of the structure prior to the intervention.

The second, carried out in March 2022, took place upon completion of the work. Repeating the same testing scheme allowed for a direct comparison between the two states, thereby reducing uncertainty associated with variations in the test setup.

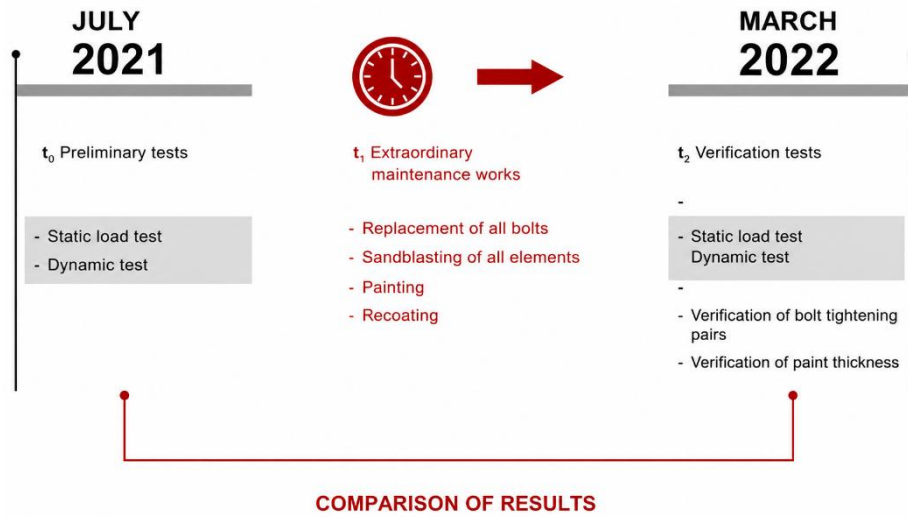


Fig. 4. Timeline of the pre- and post-intervention protocol.

Table 2. Structure of the experimental program.

Phase	Period	Key activities
Before the work begins	July 2021	Static load test; OMA/EMA dynamic characterization
Intervention	2021-2022	Bolt replacement; sandblasting; protective coating
After work	March 2022	Static load test; OMA/EMA dynamic characterization; complementary inspections

4.2 Static tests

Static load tests were conducted using a hopper car weighing approximately 44 tonnes to simulate the passage of a rail load across the deck. Six configurations were considered, encompassing both loading and unloading phases.

The most representative configurations were established by positioning the load 1.10 m from the axis of the Trapani-side bearing (measured to the front buffer) and 17.40 m from the same bearing; subsequent unloading phases allowed for the assessment of any residual deformation.

Static response was monitored using ten biaxial inclinometers installed on the top surface of the deck, oriented in both longitudinal and transverse directions. This setup enabled the reconstruction of the deformed shape of the two truss girders at mid-span and at the quarter-span points. Local deformation was evaluated using two electrical resistance strain gauges installed on the underside of the bottom chords of the downstream and upstream trusses, designated ER3 and ER6, respectively.

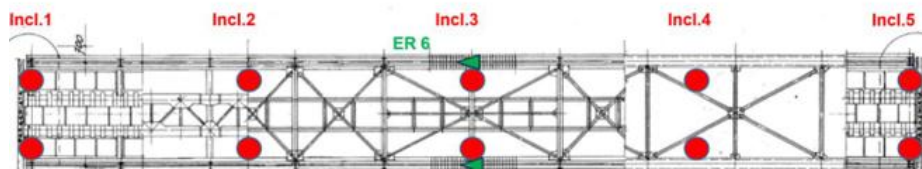
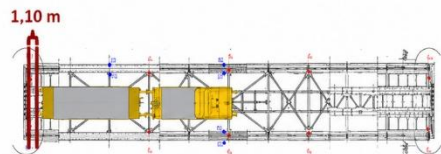


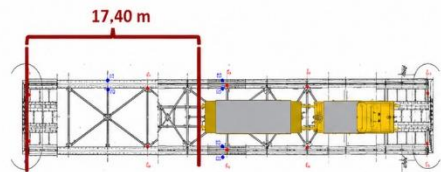
Fig. 5. Layout of static instrumentation: inclinometers and extensometers.



Load configuration no. 1 and no. 5 - pusher at 1.10 m from the support on the TP side

TP ←

→ PA



Load configuration no. 2 and no. 4 - pusher at 17.40 m from the support on the TP side

TP ←

→ PA



Fig. 6. Testing phases.

4.3 Dynamic tests (OMA/EMA)

The dynamic tests were aimed at identifying the structure's key modal parameters both before and after the intervention. Accelerometers installed at strategic points on the deck were used for data acquisition, capturing both flexural modes and any torsional components of the response. The sensors were positioned at mid-span and at the quarter-span points on both sides of the girder, measuring in both vertical and horizontal directions to obtain a representative description of the overall dynamic behavior.

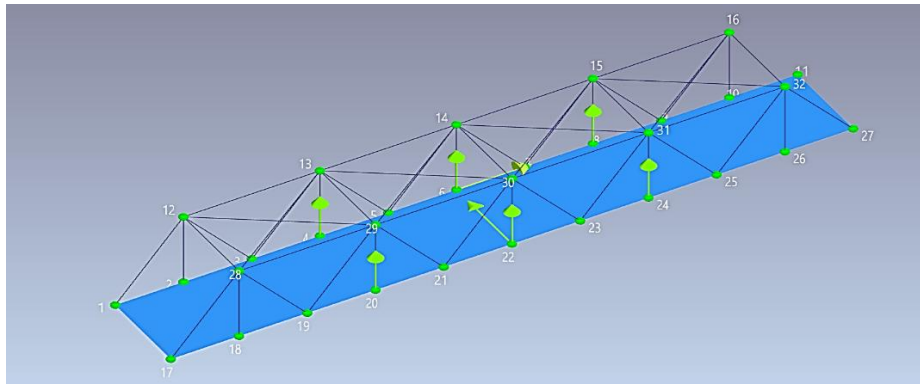


Fig. 7. Layout of the accelerometric instrumentation used for dynamic characterization.

Measurements were carried out using two complementary approaches. Operational Modal Analysis (OMA) was employed using ambient excitation—without artificial forcing—to identify the structure's natural frequencies and mode shapes [4]. Conversely, Experimental Modal Analysis (EMA), conducted using an instrumented hammer, allowed for the analysis of the free-oscillation response following a known input force, thereby supporting the estimation of damping coefficients [3].

The sampling frequency was set at 100 Hz, a value adequate for identifying the structure's first few modes. OMA data acquisition lasted long enough to ensure a stable representation of the dynamic response under ambient conditions; in this case, recordings of approximately 30 minutes were used. EMA acquisition, however, was limited to the transient phase following the impulse—lasting about 20 seconds—which was necessary for assessing damping.

The mode shapes identified in the two testing campaigns were compared using the Modal Assurance Criterion (MAC), which assesses the degree of correspondence between experimental modes before and after the intervention [6]. Values close to unity indicate good consistency between the compared mode shapes, whereas lower values may highlight differences in modal response or potential uncertainties in identification. In this instance, the MAC served as a tool to verify modal consistency and assist in interpreting the observed dynamic changes.

The combined use of OMA, EMA, and MAC analysis thus made it possible to evaluate not only changes in natural frequencies but also the stability of mode shapes and the evolution of the structure's energy-dissipation characteristics following maintenance interventions. In this way, the dynamic characterization provided a picture

complementary to the static tests—particularly useful for interpreting the influence of the restoration of bolted joints and metal surfaces on the overall behaviour of the deck.

5 Experimental results

5.1 Static response

A comparison of the deformed shapes reconstructed using inclinometers reveals a response that is generally consistent with the applied loading patterns. In the configuration with the load applied 1.10 m from the Trapani-side support, the maximum deflection decreased from 8.12 mm during the July 2021 campaign to 4.88 mm during the March 2022 campaign. Conversely, in the configuration with the load at 17.40 m, the maximum values were essentially comparable, measuring 9.61 mm before the intervention and 9.58 mm after. During the unloading phase, the residual value decreased from 1.84 mm to 1.30 mm.

Table 3. Summary comparison of maximum deflections in the pre- and post-intervention campaigns.

Test plan	Max deflection 2021 [mm]	Max deflection 2022 [mm]	Variation
Load at 1.10 m from the support on the Trapani side	8.12	4.88	-39.9%
Load at 17.40 m from the support on the Trapani side	9.61	9.58	-0.3%
Unloading after transit to Palermo	1.84	1.30	-29.3%

Strain gauge readings confirm the same pattern. The maximum strain measured at ER3 decreases from 92 $\mu\epsilon$ to 71 $\mu\epsilon$, with the associated stress dropping from 19.32 MPa to 14.91 MPa; for ER6, the strain decreases from 74 $\mu\epsilon$ to 63 $\mu\epsilon$, with stress falling from 15.54 MPa to 13.23 MPa. These values, referring solely to the added load, are low and consistent with elastic structural behaviour.

Table 4. Comparison of maximum strains and stresses measured on the bottom chords.

Sensor	$\epsilon_{\max}$ 2021 [$\mu\epsilon$]	$\sigma_{\max}$ 2021 [MPa]	$\epsilon_{\max}$ 2022 [$\mu\epsilon$]	$\sigma_{\max}$ 2022 [MPa]	Variation σ
ER3	92	19.32	71	14.91	-22.8%
ER6	74	15.54	63	13.23	-14.9%

The reduction in residual deformation upon unloading is a particularly significant indicator, as it highlights a more consistent system response under loading and unloading cycles. This improvement is not uniform across all configurations; however, a direct comparison of tests repeated using the same protocol makes it possible to distinguish global effects from those dependent on the load position.

5.2 Static response

Dynamic identification yielded four main modes, attributable to flexural and torsional forms of the deck. The pre- and post-intervention comparison reveals slight variations in natural frequencies—variations that are nonetheless significant from an interpretive standpoint. The first flexural mode shifts from 6.93 Hz to 6.24 Hz; subsequent modes show more limited variations, with frequencies of 14.01 Hz and 13.96 Hz for the second mode, 18.16 Hz and 18.02 Hz for the third, and 21.73 Hz and 21.41 Hz for the fourth.

Table 5. Key dynamic parameters identified during the two campaigns.

Mode	Type	f 2021 [Hz]	f 2022 [Hz]	Variation
1	Flexional	6.93	6.24	-9.96%
2	Torsional	14.01	13.96	-0.36%
3	Flexional	18.16	18.02	-0.77%
4	Torsional	21.73	21.41	-1.47%
Damping	-	4.51%	5.26%	+16.6%

The most significant finding concerns the damping coefficient, which increases from 4.51% to 5.26%, representing a relative increase of approximately 17%. This result indicates a change in the structure's dissipative capacity—an aspect particularly useful for interpreting the role of connections and boundary conditions following the restoration.

A summary of the dynamic results is presented in Fig. 8.

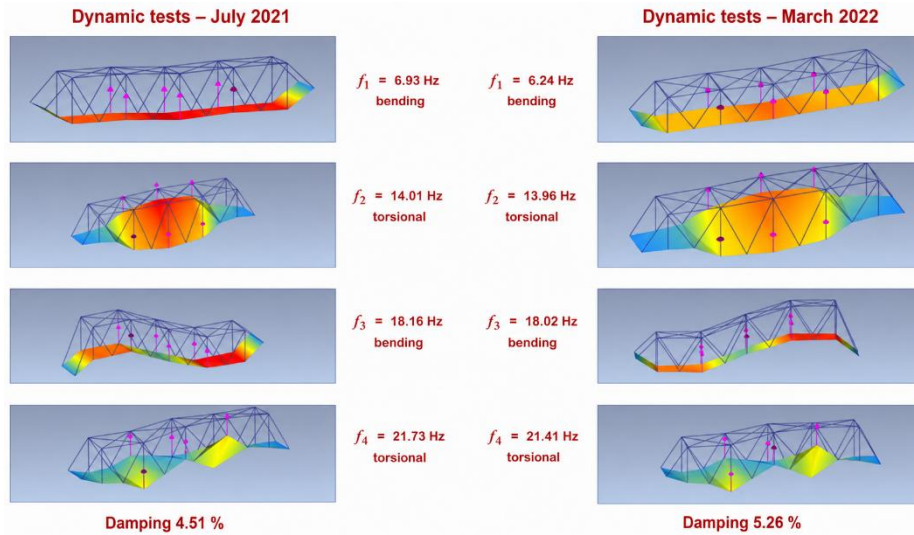


Fig. 8. Comparison of frequencies, mode shapes, and damping across testing campaigns.

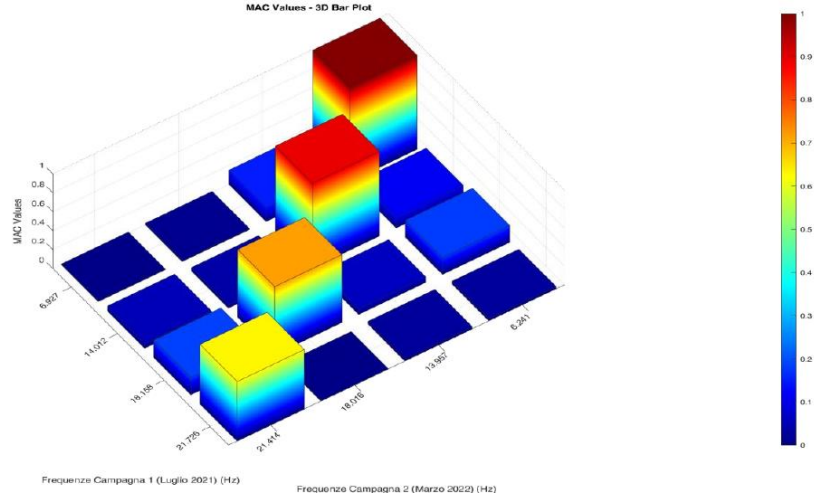


Fig. 9. Comparison of the modes identified in the two campaigns using MAC.

The MAC comparison shows the correspondence between the modal shapes identified in the two campaigns and makes it possible to verify that the observed variations are not due to an arbitrary change in modal identification. In this regard, the MAC serves as a tool for quality control of the comparison and supports the interpretation of frequency and damping variations.

5.3 Diagnostic discussion

The results indicate that the effect of maintenance interventions cannot be interpreted solely in terms of an increase in global stiffness. In the case of metal truss structures with bolted joints, corrosion can create anomalous local conditions, resulting in apparent stiffening caused by oxide formation and reduced microslip between connected elements. Bolt replacement, sandblasting, and the restoration of surface protection can therefore alter connection behavior, once again allowing for a degree of local deformability within the joints.

From this perspective, the reduction in certain natural frequencies observed after the intervention should not automatically be interpreted as a loss of performance. Instead, it can be viewed as the result of restoring boundary conditions that are less artificially stiffened by corrosion. Concurrently, the increase in damping indicates greater energy dissipation capacity in the structure, consistent with connection behavior that is more regular and less constrained by oxidative phenomena.

It must also be noted that the two testing campaigns were conducted at different times of the year: preliminary tests took place in summer, with high average temperatures, whereas post-intervention tests were performed in March, under significantly lower thermal conditions. Such differences can influence the dynamic response of a metal structure and must be taken into account when interpreting frequency data quantitatively. However, a combined analysis of frequencies, mode shapes, damping, and static results allows for a more robust assessment than relying on a single indicator.

The pre- and post-intervention comparison thus demonstrates that evaluating the effectiveness of maintenance on degraded metal structures cannot rely solely on frequency variations. It is necessary to jointly interpret deformations, residual deflections, stresses, mode shapes, damping, and modal coherence. In this regard, the integration of static and dynamic tests represents an effective tool for distinguishing between actual performance improvement, changes in boundary conditions, and local modifications induced by the interventions.

6 Conclusions

This study presented an integrated experimental procedure to evaluate the influence of corrosion and maintenance interventions on the performance of a steel truss railway bridge. The approach was applied to the Forgia Bridge, located in a marine environment and characterized by widespread corrosion, particularly concentrated at the bolted joints.

Static tests conducted before and after the intervention revealed a reduction in deflection values under certain load configurations and, notably, a decrease in residual deformation upon unloading. Strain gauge measurements showed a reduction in peak strains and stresses in the bottom chords, confirming improved elastic behavior.

Dynamic tests (OMA/EMA) provided complementary insights, showing minor variations in natural frequencies and an increase in damping of approximately 17%. This result suggests that the interventions also affected connection conditions and the structure's energy dissipation capacity—aspects that are difficult to assess through visual inspections or static testing alone.

The case study thus confirms the value of pre- and post-intervention protocols that integrate static and dynamic measurements. The identified parameters establish an experimental baseline useful for both calibrating numerical models and implementing future structural health monitoring programs.



Fig. 10. Photo of the Forgia Bridge after completion of the work.

References

- [1] De Miranda F., Steel structure railway bridges, ILVA Monograph, 1989.
- [2] UNI EN ISO 12944, Paints and varnishes - Corrosion protection of steel structures by protective paint systems, International Organization for Standardization.
- [3] Ewins D. J., Modal Testing: Theory, Practice and Application, Research Studies Press, 2000.
- [4] Brincker R., Ventura C., Introduction to Operational Modal Analysis, Wiley, 2015.
- [5] Brincker R., Zhang L., Andersen P., Modal identification from ambient responses using Frequency Domain Decomposition, Proceedings of IMAC XVIII, 2000.
- [6] Allemang R. J., The Modal Assurance Criterion - Twenty years of use and abuse, Sound and Vibration, 37(8), 2003.
- [7] Welch P. D., The use of fast Fourier transform for the estimation of power spectra, IEEE Transactions on Audio and Electroacoustics, 15(2), 1967.